AI 3000: Reinforcement Learning
Practice Problem Set 1

1 From a Markov Chain to an MDP

Consider an MDP with state space

$$\mathcal{S} = \{A, B, C\}$$

and two actions L and R available at every state. The transition matrices under the two actions are

$$P_L = \begin{pmatrix} 0.7 & 0.3 & 0 \\ 0.2 & 0.6 & 0.2 \\ 0 & 0.4 & 0.6 \end{pmatrix}, \qquad P_R = \begin{pmatrix} 0.3 & 0.7 & 0 \\ 0.1 & 0.3 & 0.6 \\ 0 & 0.5 & 0.5 \end{pmatrix}.$$

The expected one-step rewards under the two actions are

$$r_L = \begin{pmatrix} 2 \\ 4 \\ 3 \end{pmatrix}, \qquad r_R = \begin{pmatrix} 5 \\ 1 \\ 6 \end{pmatrix}.$$

1.1 Consider the stationary randomized policy

$$\pi(L \mid s) = q_s, \qquad \pi(R \mid s) = 1 - q_s.$$

Show that the transition matrix of the $\{S_n\}_{n \geq 0}$ process induced by π is obtained by mixing the corresponding rows of P_L and P_R according to the policy probabilities.

1.2 Find the values q_s, for each $s \in \mathcal{S}$, such that the policy above induces the following transition matrix for the $\{S_n\}_{n \geq 0}$ process:

$$P^\pi = \begin{pmatrix} 0.4 & 0.6 & 0 \\ 0.16 & 0.48 & 0.36 \\ 0 & 0.42 & 0.58 \end{pmatrix}.$$

1.3 Consider a two-period finite-horizon problem with total reward $(R_1 + R_2)$. Let

$$Q_1(s, a) = r(s, a)$$

denote the one-period state-action value. The two-period state-action value is given by

$$Q_2(s, a) = r(s, a) + \sum_{s' \in \mathcal{S}} P(s' \mid s, a) v_1^\pi(s'),$$

where

$$v_1^\pi(s') = \max_{a' \in \mathcal{A}} Q_1(s', a').$$

(a) Compute $v_1^\pi(s)$ for every state.

(b) Compute $Q_2(s, a)$ for all six state-action pairs.

(c) Show that the optimal deterministic policy for the two-period problem is stationary.

1.4 Following the stationary policy obtained in 1.3c and starting from $S_0 = A$, calculate the distribution of S_1 and S_2.

2 Supremum vs. Maximum in a Two-Decision MDP

Consider a finite-horizon MDP with horizon $N = 2$ and state space

$$\mathcal{S} = \{s_0, s_1, s_T\}, \qquad s_T \text{ terminal.}$$

At s_0 the agent chooses an action from the *open* interval

$$a \in (0, 1),$$

and the transition is deterministic:

$$P(s_1 \mid s_0, a) = 1, \qquad r(s_0, a, s_1) = a.$$

At s_1 the agent chooses from a finite action set $\mathcal{A}(s_1) = \{b, c\}$, both of which terminate the episode:

$$P(s_T \mid s_1, b) = 1, \quad r(s_1, b, s_T) = 0,$$

$$P(s_T \mid s_1, c) = 1, \quad r(s_1, c, s_T) = (1 - a)^2.$$

Recall from lecture: $v_N^\pi(s)$ is the expected N-step return of $\pi \in \Pi^{HR}$ from s, and

$$v_N^\star(s) = \sup_{\pi \in \Pi^{HR}} v_N^\pi(s), \qquad \pi^\star \in \operatorname*{argsup}_{\pi \in \Pi^{HR}} v_N^\pi(s).$$

2.1 **Computing $v_N^\star(s_0)$.**

(a) For fixed $a \in (0, 1)$, determine the optimal action at s_1.

(b) Assuming the agent follows the optimal action deterministically at s_1, write $v_1^\pi(s_1)$ and then $v_2^\pi(s_0)$ as an explicit function of a alone. Call this function $g(a)$.

(c) Determine whether g is monotone on $(0, 1)$. Where is its minimum? What does this imply about where a maximizer *would* have to sit if the domain were closed?

(d) Show that

$$v_2^\star(s_0) = \sup_{\pi \in \Pi^{HR}} v_2^\pi(s_0) = \sup_{(0,1)} g(a) = 1.$$

2.2 **No optimal policy exists.**

In this part, we will work through a series of logical steps to formally prove that no $\pi^\star \in \Pi^{HR}$ attains $v_2^\star(s_0)$, i.e. that

$$v_2^\pi(s_0) < v_2^\star(s_0) \qquad \text{for every } \pi \in \Pi^{HR}.$$

(a) Fix $a \in (0, 1)$ and let π_a denote the deterministic policy that plays a at s_0 (followed by the optimal s_1-action from 2.1. Compute the value of

$$v_2^\star(s_0) - v_2^{\pi_a}(s_0),$$

and show that it factors as a product of two strictly positive terms.

(b) Using 2.2a, show that $v_2^{\pi_a}(s_0) < v_2^\star(s_0)$ for *every* $a \in (0,1)$ — i.e. no deterministic policy attains $v_2^\star(s_0)$. Then exhibit two sequences, $(a_n) \to 0^+$ and $(a'_n) \to 1^-$, along which $v_2^{\pi_{a_n}}(s_0) \to v_2^\star(s_0)$ and $v_2^{\pi_{a'_n}}(s_0) \to v_2^\star(s_0)$ respectively. What does it mean, geometrically, that the bound is approached from *both* ends of the interval rather than just one?

(c) Now let $\pi \in \Pi^{HR}$ be arbitrary. Since the history up to s_0 is trivial, π reduces to choosing $A \sim \mu$ at s_0 for some distribution μ on $(0,1)$, followed by the optimal s_1-action. Show that

$$v_2^\pi(s_0) = \mathbb{E}_{A\sim\mu}\Big[g(A)\Big], \qquad \text{where } g(a) := v_2^{\pi_a}(s_0).$$

Using the strict inequality from 2.2b (which holds for *every* a in the support of μ), argue that

$$\mathbb{E}_{A\sim\mu}[g(A)] < v_2^\star(s_0)$$

for every distribution μ on $(0,1)$ — degenerate or not. Conclude that $\mathbb{P}(v_2^\pi(s_0) = v_2^\star(s_0))$ cannot hold for any $\pi \in \Pi^{HR}$, and hence no optimal policy exists.

2.3 **An ϵ-optimal policy.**

For $\epsilon \in (0,1)$, construct an explicit deterministic policy π^ϵ (i.e., an explicit choice $a_\epsilon \in (0,1)$, together with the optimal s_1-action from 2.1 satisfying

$$v_2^{\pi^\epsilon}(s_0) > v_2^\star(s_0) - \epsilon.$$

Use the factored expression from 2.2a of 2.2 to justify your choice of a_ϵ with a short inequality, not just a limiting argument.